

6 Fitting models

$$6.1 \quad \lambda(\phi) = \sum_{i=1}^n \ell_i = \sum_{i=1}^n (f(x_i, \phi) - y_i)^2 = \sum_{i=1}^n (\phi_0 + \phi_1 x_i - y_i)^2$$

$$\frac{\partial \ell_i}{\partial \phi_0} = 2(\phi_0 + \phi_1 x_i - y_i) \cdot 1$$

$$\frac{\partial \ell_i}{\partial \phi_1} = 2(\phi_0 + \phi_1 x_i - y_i) x_i$$

$$6.2 \quad H(\phi) = \begin{pmatrix} \frac{\partial^2 \lambda}{\partial \phi_0^2} & \frac{\partial^2 \lambda}{\partial \phi_0 \partial \phi_1} \\ \frac{\partial^2 \lambda}{\partial \phi_1 \partial \phi_0} & \frac{\partial^2 \lambda}{\partial \phi_1^2} \end{pmatrix} \quad \begin{array}{l} H(\phi) \text{ positive } \forall \phi \in H \\ \Downarrow \\ \text{Surface } \phi \text{ convex} \\ \Downarrow \\ \text{Surface has unique minimum} \end{array}$$

$$\frac{\partial^2 \ell_i}{\partial \phi_0 \partial \phi_1} = \frac{\partial}{\partial \phi_1} 2(\phi_0 + \phi_1 x_i - y_i) = 2x_i$$

$$\frac{\partial^2 \ell_i}{\partial \phi_1 \partial \phi_0} = \frac{\partial}{\partial \phi_0} 2x_i \phi_0 + 2\phi_1 x_i^2 - 2y_i x_i = 2x_i$$

$$\frac{\partial^2 \ell_i}{\partial \phi_0^2} = 2$$

$$\frac{\partial^2 \ell_i}{\partial \phi_1^2} = 2x_i^2$$

$$\text{Thus } H = 2 \begin{pmatrix} 1 & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}$$

$\Rightarrow$ convex!

$$\text{trace}(H) = (1 + \sum_{i=1}^n x_i^2) 2 > 0$$

$$\det(H) = 4 \left(1 \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 \right) = 4 \sum_{i=1}^n (x_i - \bar{x})^2$$

as $1 > 1$ $\det(H) > 0$ where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$$6.3 \quad L(\Phi) = \sum_{i=1}^I (p(x_i, \Phi) - y_i^2) \quad =: a$$

$$= \sum_{i=1}^I \left(\underbrace{\sin(\Phi_0 + \Phi_1 x_i)}_{=: p} \cdot \underbrace{\exp\left[-\frac{(\Phi_0 + 0.06 \cdot \Phi_1 x_i)^2}{32}\right]}_{=: g} - y_i \right)^2$$

$$\frac{\partial L}{\partial \Phi_0} = \sum_{i=1}^I \left(2a \cdot \left(\frac{\partial p}{\partial \Phi_0} g + \frac{\partial g}{\partial \Phi_0} p \right) \right)$$

$$\frac{\partial p}{\partial \Phi_0} = \cos(\Phi_0 + \Phi_1 x_i) \cdot 1$$

$$\frac{\partial g}{\partial \Phi_0} = \exp\left[-\frac{(\Phi_0 + 0.06 \cdot \Phi_1 x_i)^2}{32}\right] \cdot \left(-\frac{1}{16}\right) (\Phi_0 + 0.06 \cdot \Phi_1 x_i) \cdot 1$$

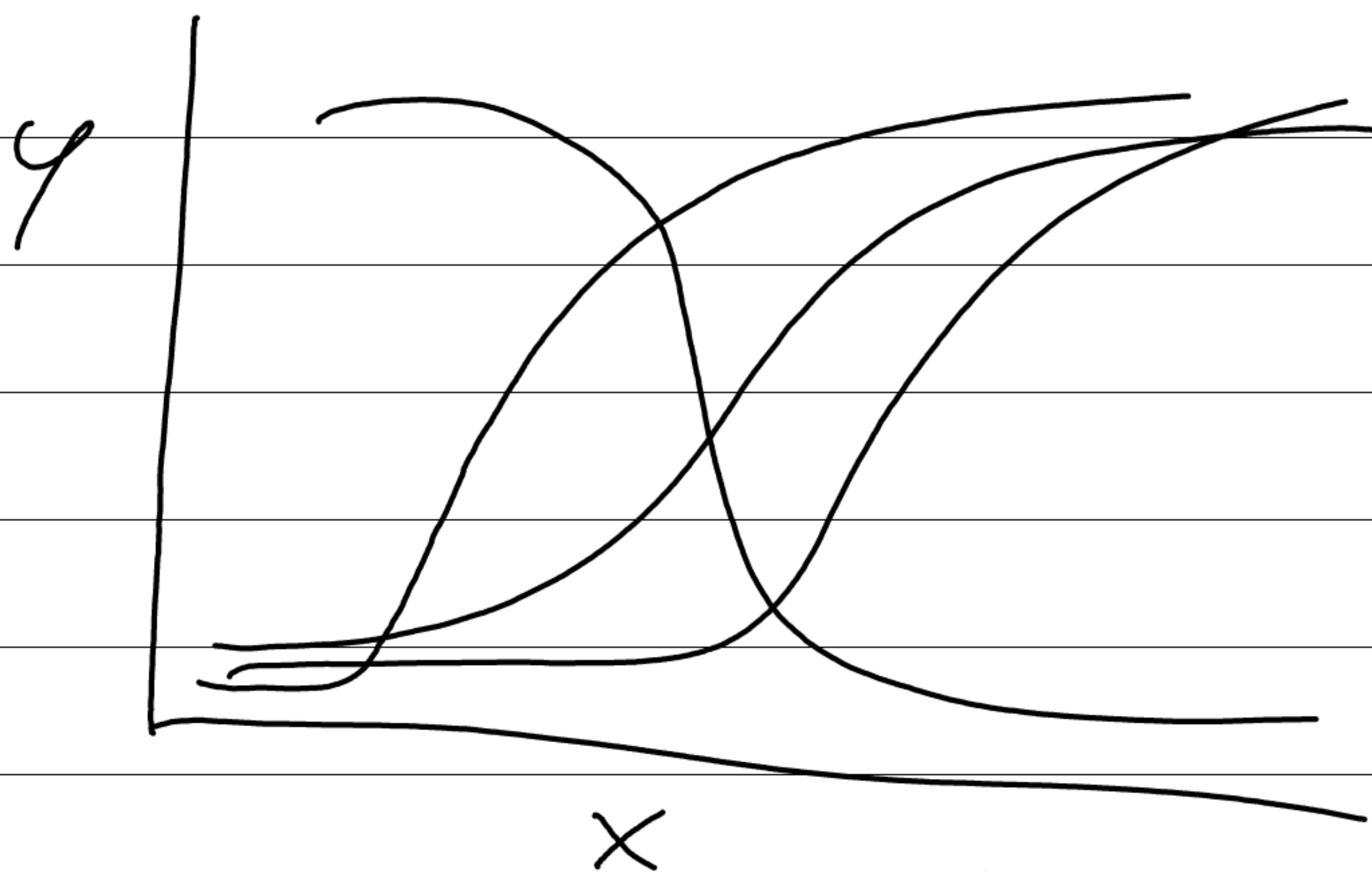
$$\frac{\partial L}{\partial \Phi_1} = \sum_{i=1}^I \left(2a \cdot \left(\frac{\partial p}{\partial \Phi_1} g + \frac{\partial g}{\partial \Phi_1} p \right) \right)$$

$$\frac{\partial p}{\partial \Phi_1} = \cos(\Phi_0 + \Phi_1 x_i) \cdot x_i$$

$$\frac{\partial g}{\partial \Phi_1} = \exp\left[-\frac{(\Phi_0 + 0.06 \cdot \Phi_1 x_i)^2}{32}\right] \cdot \left(-\frac{1}{16}\right) \cdot 0.06 x_i$$



6.4
(i)



dec ϕ_0 repr.
a shift
and ϕ_1 slope/
mirror

(ii) e.g. cross-entropy loss $h(\phi) = -\sum_{i=1}^1 \log [\text{softmax}_i [h(x_i, \phi)]]$

(iii) let $z_i := \phi_0 + \phi_1 x$, $p_i = \text{sig}(z_i)$.

$$\text{as } \frac{d}{dz} \log(1 + \exp(z)) = \text{sig}(z)$$

$$\frac{\partial h}{\partial \phi_0} = \sum_{i=1}^1 (p_i - y_i) \text{ and } \frac{\partial h}{\partial \phi_1} = \sum_{i=1}^1 (p_i - y_i) x_i$$

(iv) looks like half circles ϕ_0 ϕ_1 $\rightarrow$ minima

(v) Yes. We could look at the Hessian and
prove that its eigenvalues are strictly positive.
Then it is convex.

$$6.5 \quad J(\phi) = \sum_{i=1}^n (f(x_i, \phi) - y_i)^2 \quad \text{ReLU}(x) = \max(0, x)$$

$$\frac{d \text{ReLU}(x)}{dx} = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}, \text{ it's undefined at } 0?$$

$$= I(x > 0)$$

$$\frac{\partial J}{\partial \phi_j} = \frac{\partial}{\partial \phi_j} \sum_{i=1}^n (f(x_i, \phi) - y_i)^2$$

$$= 2 \sum_{i=1}^n (f(x_i, \phi) - y_i) \cdot \frac{\partial}{\partial \phi_j} f(x_i, \phi)$$

$$\text{with } f(x, \phi) = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11} x] + \phi_2 a[\theta_{20} + \theta_{21} x] + \phi_3 a[\theta_{30} + \theta_{31} x]$$

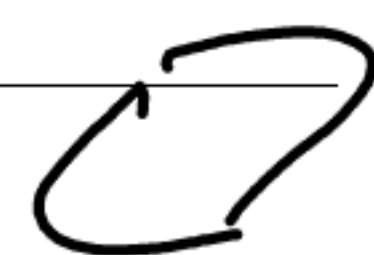
thus:

$$\frac{\partial J}{\partial \phi_0} = 1 \quad \frac{\partial J}{\partial \phi_1} = a[\theta_{10} + \theta_{11} x_i] \quad \frac{\partial J}{\partial \phi_2} = a[\theta_{20} + \theta_{21} x_i]$$

$$\frac{\partial J}{\partial \theta_{10}} = \phi_1 I[\theta_{10} + \theta_{11} x_i > 0] \quad \frac{\partial J}{\partial \theta_{11}} = \frac{\partial J}{\partial \theta_{10}} \cdot x_i$$

$$\frac{\partial J}{\partial \theta_{20}} = \phi_2 I[\theta_{20} + \theta_{21} x_i > 0] \quad \frac{\partial J}{\partial \theta_{21}} = \frac{\partial J}{\partial \theta_{20}} \cdot x_i$$

$$\frac{\partial J}{\partial \theta_{30}} = \phi_3 I[\theta_{30} + \theta_{31} x_i > 0] \quad \frac{\partial J}{\partial \theta_{31}} = \frac{\partial J}{\partial \theta_{30}} \cdot x_i$$



6.6 a) not convex, as local minima exist, 1: local, 2: global, 3: local

b) convex, as only one global minima, 4: neither, 5: global

c) not convex as only one global minima, 6: global, 7: neither (inflection)

6.7

$L(\phi)$ has a narrow valley: $H = \nabla^2 L$ has

eigenvalues with $\lambda_{\max} \gg \lambda_{\min}$.

This is because with a fixed step size α , each step 'overshoots' the valley floor; at the next step the gradient's signs flip, producing the back and forth $\leadsto$ solution: momentum / reduce α

6.8 Yes, by averaging over the local minima.

6.9 batch size: 20 (how many examples / update step)

iteration: 1000 (one update using one batch)

epoch: one full pass through the dataset: epochs = $\frac{\text{iterations}}{(\text{examples}) / (\text{batch size})}$

dataset: 100 examples

$$\text{epochs} = \frac{1000}{100/20} = \frac{1000}{5} = 200$$

$$6.10 \quad m_{t+1} \leftarrow \beta m_t + (1-\beta) \sum_{i \in B_t} \frac{\partial \ell_i(\phi_t)}{\partial \phi}$$

$$\phi_{t+1} \leftarrow \phi_t - \alpha m_{t+1}$$

$$(\text{let } g_t = \sum_{i \in B_t} \frac{\partial \ell_i(\phi_t)}{\partial \phi})$$

$$\begin{aligned} & \text{as } t \rightarrow \infty, \\ & (1-\beta) \beta^k \text{ goes to } 0 \\ & \circ \quad w_0 = 1-\beta \end{aligned}$$

$$m_{t+1} = \beta (\beta m_{t-1} + (1-\beta) g_{t-1}) + (1-\beta) g_t$$

$$= \beta^2 m_{t-1} + \beta(1-\beta) g_{t-1} + (1-\beta) g_t$$

$$= \beta^2 (\beta m_{t-2} + (1-\beta) g_{t-2}) + \beta(1-\beta) g_{t-1} + (1-\beta) g_t$$

$$= \beta^3 m_{t-2} + \beta^2(1-\beta) g_{t-2} + \beta(1-\beta) g_{t-1} + (1-\beta) g_t$$

$$= (1-\beta) \sum_{k=0}^{\infty} \beta^k g_{t-k} \Rightarrow w_k = (1-\beta) \beta^k \quad \square$$

$$6.11 \quad H = \begin{pmatrix} \frac{\partial^2 L}{\partial \phi_1^2} & \frac{\partial^2 L}{\partial \phi_1 \partial \phi_2} & \dots \\ \vdots & \ddots & \ddots \\ \frac{\partial^2 L}{\partial \phi_1 \dots \partial \phi_{999999}} & \dots & \frac{\partial^2 L}{\partial \phi_{999999}^2} \end{pmatrix} \leadsto \dim(H) = 1000000 \times 1000000$$

(only ϕ for this example)